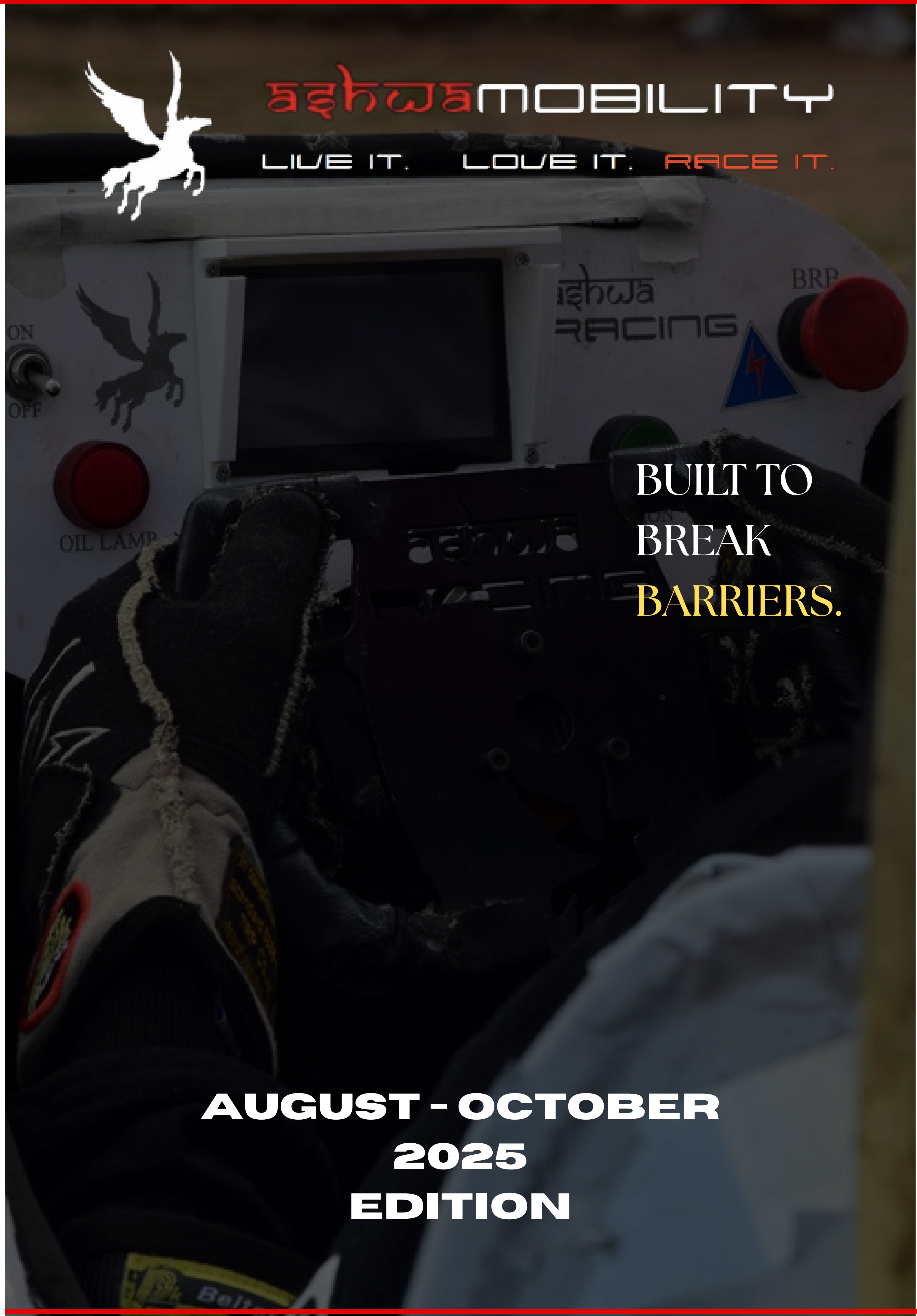


ashwa MOBILITY

LIVE IT. LOVE IT. RACE IT.

BUILT TO
BREAK
BARRIERS.

AUGUST - OCTOBER
2025
EDITION

CONTENTS

“VISION AND MISSION”

GUIDING PURPOSE,
INSPIRING FUTURE.

MISSION AND VISION

VISION

- To be the national leader in all categories of Formula Student, to push the limits of engineering and compete in events at the highest level.

- To foster student mobility solutions in the future by developing sustainable prototypes across various domains.

MISSION

- Prepare students for the industry by designing and building world-class vehicles while facilitating industry interaction.

- Showcase India's engineering expertise in mobility on a global stage and provide talent opportunities for the industry.

- Develop cost-effective, sustainable transportation solutions for the masses.

“

STUDENT INDUCTION PROGRAM

”

IGNITING FRESH TALENT.

STUDENT INDUCTION PROGRAM

The Ashwa Student Induction Program for the freshers was a high-octane success, where the freshers from Batch of 29’ were taken deep inside the world of speed, science, and strategy that is Formula Student. Entirely planned and conducted by the focused Batch of '28, the day-long event set the pace for the season ahead.

1. Beyond the Apex: Building the Dream
Senior team members introduced the freshers to the anatomy of a race car in a deep-dive session. The lecture covered all technical pillars dealing with Formula Student, from the engine and brakes to electrical systems, aerodynamics, and chassis. This session allowed freshers to understand the basics required to choose their path in Ashwa.

2. Grand Prix Gambit: Race Strategy
This competition presented the F1 race strategies and engineering decisions as a challenge. The competition was indeed a test of strategic thinking, eliciting from these very sharp minds what would come closest to that thinking evident in motorsports.

3. Fasten and Furious: Hands-On Assembly
Freshers were given an explanation on the working of the uprights of the front wheels and a chance to compete to assemble the parts precisely under a time limit. This exercise provided immediate exposure to the accuracy and teamwork required in the workshop.

4. Overdrive: The Gaming Station
The gaming station powered by Red Bull provided a high-octane break. Participants jumped into racing simulators for high-speed competition, nurturing a spirit of friendly rivalry and passion for the track.

5. Pitstop Challenge
An all-day, hands-on experience where freshers got to sample the type of high-intensity teamwork required to execute a pit stop. The emphasis was on coordination, efficiency, and speed; the essential hallmarks of any successful racing team.

“

BUSINESS PLAN PRESENTATION

”

DESIGN. COST. BUSINESS—ALIGNED

BUSINESS PLAN PRESENTATION

Ashwa Racing stepped into its annual Business Plan Presentation (BPP) cycle with a clear objective: rebuild the process, correct last year’s mistakes, and deliver a pitch that reflects strong technical thinking backed with real commercial logic. What followed was a structured, collaborative, and tightly-executed effort that shaped this year’s BPP narrative.

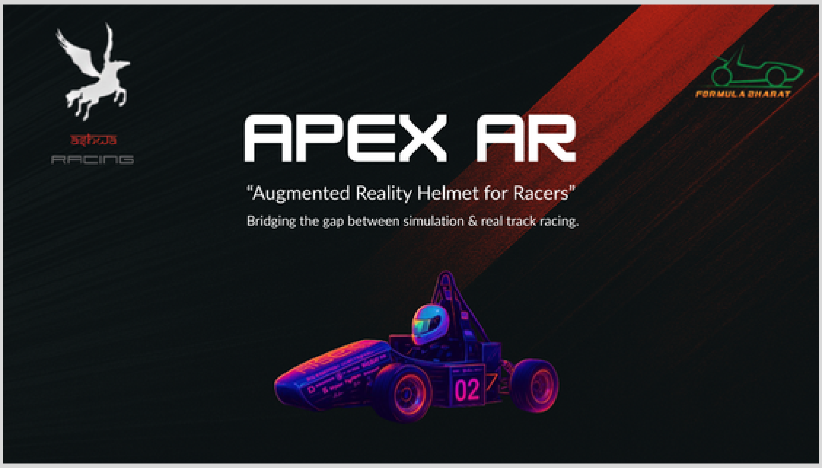
A Fresh Start grounded in Lessons Learned
The cycle began with an honest assessment of last year’s performance. The team recognised that the previous concept didn’t match market needs, research was shallow, and internal organisation was inconsistent. Work sessions lacked structure, and the final pitch felt scattered.
Acknowledging these failures wasn’t just a review, it became the foundation for how this year’s BPP would be rebuilt: clearer planning, deeper research, stronger coordination, and a more disciplined approach to storytelling.

Building This Year’s Approach
With that mindset, the team kicked off idea generation. Eight concepts were pitched, debated, and filtered down through multiple discussions. The helmet-centric concept introduced by Samith emerged as the front-runner, not because it was convenient, but because it checked every strategic requirement:

- A defined, growing market
- Straightforward research and validation
- A direct link between technical features and real-world value

The team immediately began exploring user pain points, safety gaps, and industry demand. This early research set the tone—not jumping to solutions, but grounding the storyline in a clear, validated problem.

Building the Narrative Before the Numbers
Unlike the previous year, the team built the entire pitch narrative before touching financials. This ensured that technical reasoning, market need, value proposition, and product positioning were aligned right from the start.



Parallel roles were defined to keep the structure tight:

- Aditeya built the technical justification.
- Vidya developed the VR component for visual clarity.
- Others focused on refining the storyline, ensuring every idea connected logically to the next.

Financials Under Pressure

Once the concept was stable, the team moved into a three-day financial sprint to transform the pitch from a conceptual idea into a commercially viable plan with tangible numbers:

- Nadeem was brought in as CFO for the cycle.
- Working closely with Pranav, he built costing, pricing strategies, revenue models, and growth projections.
- The team mapped production constraints, distribution challenges, and scalability opportunities.

Trial Runs, Timing, and Refinement

With content and financials ready, the team shifted into execution mode.
About 15 trial runs were conducted to perfect pacing, delivery, and confidence.

- Damian and Ananth supported with deck refinement and structure.
- Ranjith handled timing sessions, ensuring the team stayed within limits without compromising clarity.

By the final rehearsal, the pitch felt cohesive, each section flowing naturally into the next, and the presenters carried a unified narrative.

Event Day: Delivery & Feedback

On presentation day, the team delivered a structured, confident, and well-paced pitch. Judges appreciated the clarity in the storyline and the balance between engineering depth and business reasoning.
The feedback focused on two critical improvements:

- Expand beyond assembling in India—demonstrate manufacturing feasibility.
- Correct the market capture curve—real growth is exponential, not linear.

These insights weren't setbacks—they became direction for the next iteration.

A Stronger Foundation Moving Forward

This year's BPP cycle marked a clear step up for the team:

- Smarter concept selection
- Better coordination across subsystems
- Faster and more accurate financial modelling
- Stronger narrative flow
- Thorough preparation and practice

Our Journey to the Finals

The team received wonderful news of having made it to the Top 5 teams of the BPP Finals in the combustion category. The finals will be held on-site at the Formula Bharat competition in 2026 at Kari Motor Speedway.

FORGING THE CHASSIS

FROM BLUEPRINT
TO BUILD

FORGING THE CHASSIS

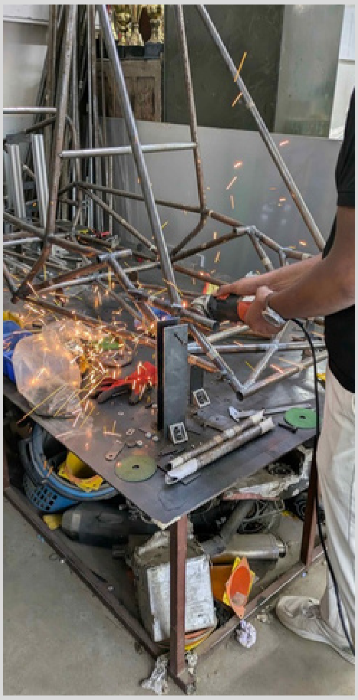
Every Formula Student car begins as an idea on CAD, but the final product? Unified in steel.

From procurement to final welds, the chassis defines stiffness, safety, and packaging for every system that it accommodates. At Ashwa Racing, the 2026 build of combustion vehicle for Formula Bharat saw a few changes in manufacturing style, the use of sheet metal fixtures for the first time.

The chassis uses AISI 1018 grade Mild Steel Tubes (both round and square tubes). All tubes underwent laser cutting and notching to fit the required lengths and profiles across different parts of the chassis. The importance of achieving precise notching is to get perfect tube-to-tube fitment, minimized gaps, and reduced weld distortion.

Now, to hold the tubes in place for welding, we deployed fixtures and a baseplate. MS baseplate was laser cut to serve as the rigid foundation for the entire structure, preventing twist or sag during welding. Every fixture, tube and clamp must be assembled onto the baseplate to bring the chassis to life. MS sheets were laser cut to form the modular fixtures that fasten and locate each tube and clamp that must be welded on the chassis.

Once the skeleton was aligned, this assembly was ready for welding. TIG welding provides the accuracy and heat control needed to weld thin MS1018 chassis tubes without distortion, delivering clean, structurally consistent, and inspection-ready joints. The process was carried out in two stages:
-Tacking: Small tacks were applied at all joints with the chassis still supported by the fixtures. This prevented the frame from losing its designed geometry.
-Seaming: After cross-checking position and alignment of all tubes, seam welds were applied to fill space and complete the welding and avoid warping.



A chassis isn’t complete without the mounting points that hold every major subsystem together. Dedicated laser-cut clamps were fixed and welded for example:

- Drivetrain components including the differential and chain tensioner
- Steering column and rack mounts
- Front and rear suspension inboards
- Engine mounts

With the welded chassis complete, the next step was to transform the skeleton into a vehicle assembly. This constituted mounting the engine first, followed by the firewall and fuel tank. Each subsystem is fixed on the chassis to test accuracy and performance of the full integration of components.

“

WHY THE IMU MATTERS

”

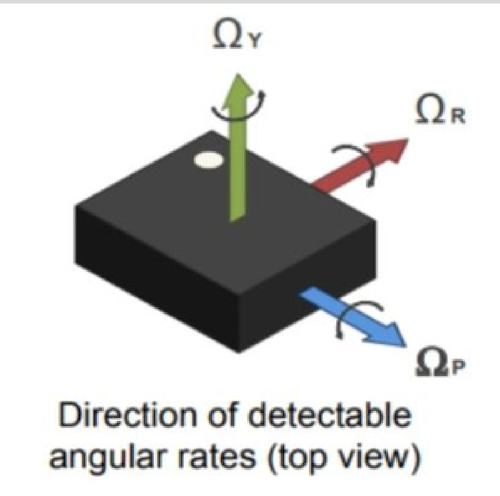
MOTION MEETS PRECISION

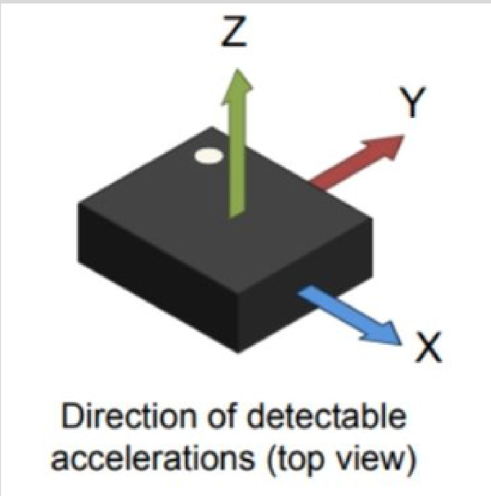
RACING IS PHYSICS; THE IMU GIVES YOU THE CONSTANTS

Ever wondered how your car still knows its position when GPS signals drop, like in tunnels or dense forests? The control unit doesn't go blind. It relies on an Inertial Measurement Unit (IMU), which tracks motion even when satellites can't. In our 26-car prototype, Ashwa Racing uses an Inertial Navigation System (INS) that blends data from both the IMU and GNSS (Global Navigation Satellite Systems) to maintain accurate positioning.

Now that we understand the need for IMU, the next question to decipher is what the IMU itself is. From its need, we get the idea that the IMU is a device that can indicate its position. It is an electronic device that makes use of sensor fusion technology to achieve the same. Which sensors does it fuse? Depending upon the IMU you're using, it can be a 6-axis or 9-axis IMU. The 6-axis consists of an accelerometer and a gyroscope. This article will be related to the 6-axis IMU, ASM330LHB from STMicroelectronics.

ASM330LHB has two components, a 3-axis accelerometer and a 3-axis gyroscope. The accelerometer uses X, Y, and Z axes for measuring linear acceleration in terms of g-forces. So if the IMU is placed flat on a table, it reads the amount of g force acting the negative Z direction. If the IMU is tilted about the Y axis, then the corresponding g forces in the X and Z directions will be obtained. Similarly, the gyroscope reads the angular acceleration along its Roll, Pitch, and Yaw axes in the unit of degrees per second (dps).





IMU accelerometer and gyrometer diagrams

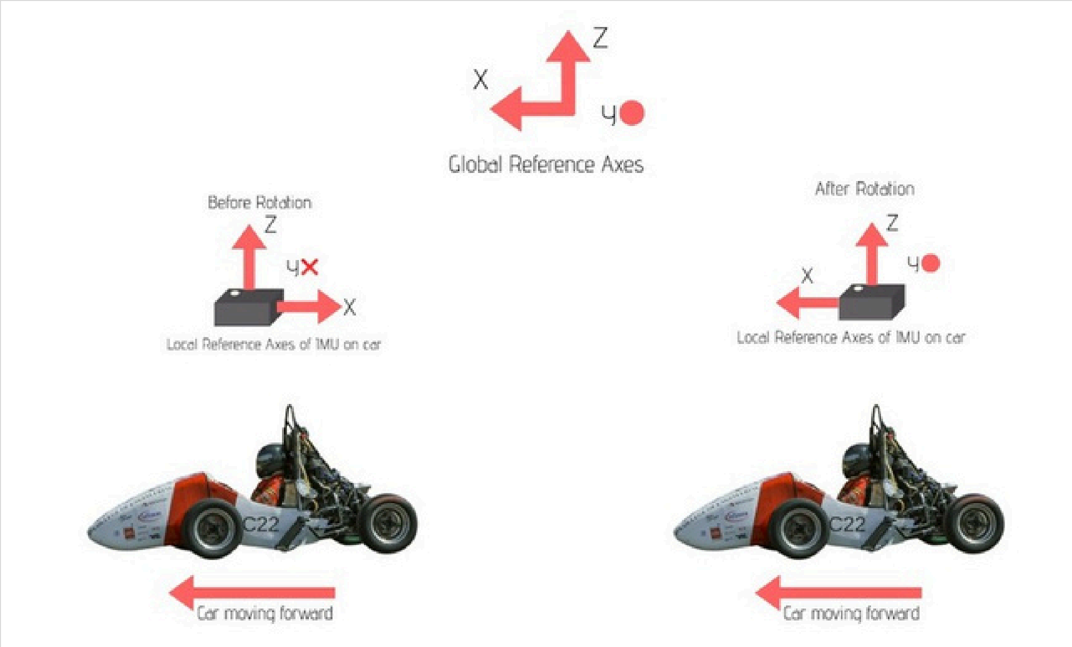


Fig. Access alignment of IMU with global reference

However, chances are, the axes (X, Y, Z, and Roll, Pitch, Yaw) are local to your IMU and may not align with the Global axes. For example, the local X of your IMU might be oriented opposite to the Global X, i.e, if your car is moving in the positive X direction, the IMU will read the data along the direction of negative X in the global reference. One way to fix this is to manually orient the local reference of the IMU to match the global reference. However, there is also a software solution to this. We make use of vector and matrix multiplication for the same. The ASM330 is interfaced over I3C to an STM32 MCU, and the math is carried out using the CMSIS-DSP library. The vectors $[X\ Y\ Z]^T$ and $[\text{Roll Pitch Yaw}]^T$ are taken such that the IMU's local reference values will be reflected in the variables. Using standard rotation matrices, we can set the angles at which the IMU is to be rotated in order to align itself with the global reference. These matrices, when multiplied with the vectors, result in vectors displaying values in accordance with the global reference, and Voila! The IMU is now calibrated and ready to use. So if your car is moving forward, the IMU shows forward.

The calibrated IMU data along with GNSS and other sensor data flow through the DAQ system, where the slave boards power the sensors and send measurements to the master for display, logging, and wireless transmission. This setup embodies Ashwa Racing's spirit—bringing together sensors, control systems, telemetry, and teamwork. Whether you're into speed, safety, or systems, there's a place to grow for everyone.

Live it, Love it, Race it.

— Team Ashwa Racing

Signed,
Anarghya Hatti,
Electrical and Testing

“ **ELECTRICAL EXCELLENCE** ”

IGNITING FRESH
TALENT.

ELECTRIC EXCELLENCE: ASHWA'S HARDWARE HUSTLE LEADS THE CHARGE

The month of October has been a period of significant technical progress and key industry collaborations for **Ashwa Racing.** As part of our mission to design and build cutting-edge electric and autonomous mobility systems, our teams across Electronics, Firmware, Data Acquisition, and Embedded Systems made major strides in both development and testing.

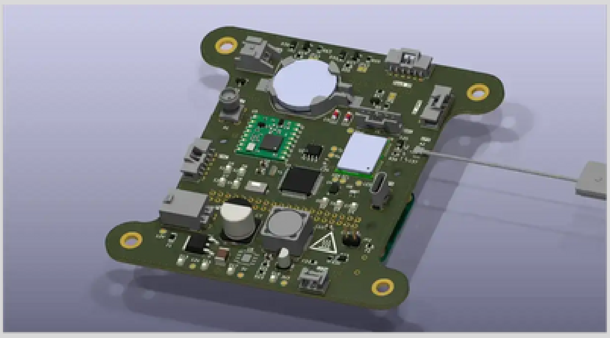
Forging Professional Partnerships

The trust and support of our industry sponsors is what helps save AMF precious resources like time and money and make our prototype possible. We secured crucial sponsorships such as **PC Process and LionCircuits.**
Our members delved into the complex world of professional manufacturing, ensuring every component meets the uncompromising standards of competitive electric racing. This involved using state-of-the-art technology to produce precise PCBs, stencils, and multi-layered boards.

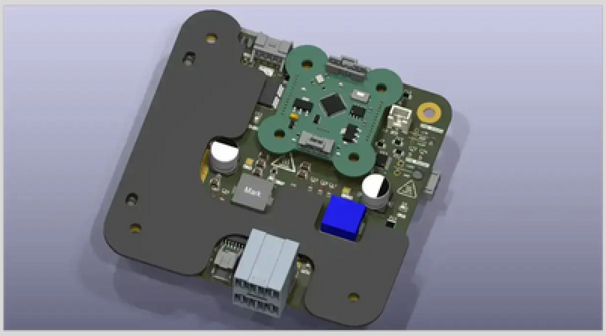
After several constructive and highly insightful technical discussions, we are proud to announce LionCircuits as an official sponsor of Ashwa Racing for the 2026 race season. We extend our sincere gratitude to their team for this collaboration. **LionCircuits** sponsored the manufacturing of the boards with their reliable fabrication quality that strengthened our general electronics pipeline, from communication modules to safety-critical logic boards. With consistent delivery timelines and thorough testing, LionCircuits helped streamline our overall hardware development beyond just the DAQ systems.

Ashwa Racing proudly acknowledges our PCB manufacturing sponsor **PC Process**, whose precise and fast fabrication made our **CAN Watchdog Board** used to constantly monitors CAN activity and triggers if any communication fails, **DAQ Front Slave Board and DAQ Side Slave Boards** that are dedicated DAQ modules placed on the vehicle front/side pods respectively to collect localized sensor data. Their high-quality processes ensured reliable multilayer builds and clean routing—crucial for the harsh electrical demands of an FSAE vehicle. Their support didn't just give us PCBs; it strengthened the core of our electrical system.

JLCPCB manufactured–**DAQ Master Board, Power Unit and Gateway** along with stencils for all PCBs.



DAQ Master



Power Unit

The **DAQ Master (Data Acquisition System)** collects and organizes all the critical sensor data from across the car—such as temperatures, wheel speeds, suspension travel, driver inputs, and battery information. It serves as the central hub for monitoring vehicle performance, logging data for analysis, and supporting real-time telemetry and diagnostics during testing and events.

The **Power Unit** manages and protects the car's entire low-voltage electrical system and performs accurate current sensing for fault detection and early diagnosis of issues. It distributes power from the GLV battery to all essential electronics—ECUs, pumps, fans, sensors, lights, and safety systems—while providing fusing, current protection, and controlled switching. The PDB ensures every subsystem receives clean, reliable power for safe and consistent operation.

Optimizing Sourcing and Testing for Better Builds

The procurement process was a professional undertaking, managed by our team to ensure quality and traceability.

- **Element14** served as the main vendor for passive components.
- **Specialized Components** were directly sourced from Texas Instruments, STMicroelectronics, Analog Devices, LittelFuse, Vishay and TEConnectivity.

- **Automotive Grade Logic:** Critically, all components were procured with **AEC-Q100/101 Grading**—These ratings ensure that the electronics can withstand wide temperature swings, continuous vibration, electrical noise, mechanical stress, and long operational lifetimes—conditions far harsher than standard consumer applications. By choosing AEC-Q100/101-qualified parts, the system ensures long-term reliability, safety, and predictable performance even under the demanding realities of on-road operation.

Rigorous Testing and System Integration

Our Testing Team conducted extensive subsystem and sensor validation this month.

- **Data Acquisition & Telemetry**: We conducted comprehensive testing of the newly developed data acquisition system on the existing chassis for validation of complex designs.
- Our team prototyped the vehicle's CAN FD data bus with the newly assembled DAQ Master and the DAQ slaves, and validated its backward compatibility with the legacy CAN2.0 bus utilised by the ECU on the combustion prototype.

Software Development: Code for the Circuit

Our members utilized **STM32CubeIDE** and **Python** to develop custom firmware, serving as the "brains" for a multitude of custom boards: DAQ Master, IO boards, Power Unit and Vehicle Control Unit (VCU). Significant refinements were implemented to support improved data acquisition speeds, better power management, and robust CAN-FD communication.

With procurement simplified through strategic sourcing, sponsorships, and automotive-grade standards, the final assembly phase is now underway. Boards are split across multiple substations for faster assembly, and each PCB uses only SMD components for a compact and robust design. Together, these steps bring the project to a unified, high-quality finish.

As we move into the next phase of development, Ashwa Racing continues to push forward in all directions of our electric subsystem. Our progress is driven by the collaborative power of our team and the invaluable support of our partners.
Thank you for following our journey. More exciting updates await in the coming months.

Live it, Love it, Race it.

— Team Ashwa Racing

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